HOVER: Part II of the Scientific Proposal

1. Preliminary results and further background

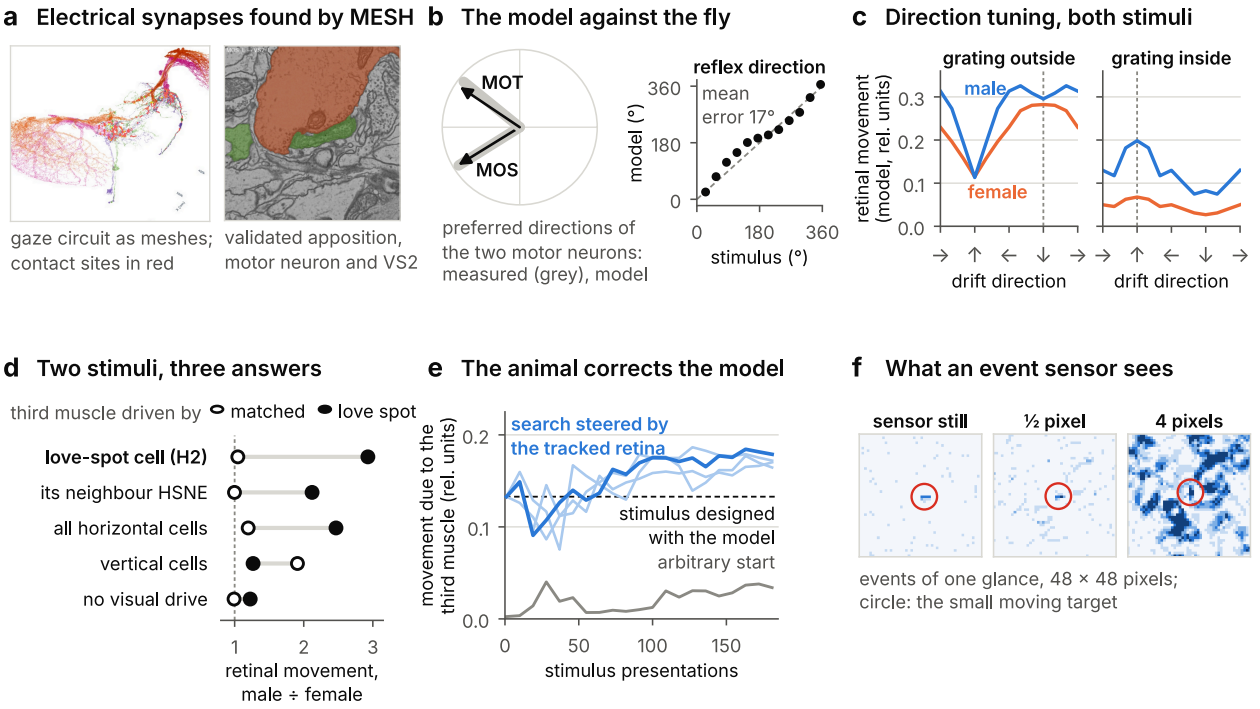


Figure B1. *Evidence behind the preliminary results (Part I, Fig. 2).* (a) MESH: the gaze circuit as neuron meshes with candidate contact sites, and a validated membrane apposition (FlyWire, 8 nm) [14,41]. (b) The model against *Drosophila* [14]. (c) Predicted retinal movement of male and female hoverflies against the drift direction of a square-wave grating outside or inside the love spot (dashed lines: the two stimuli of Part I). (d) Male-to-female ratio of that movement for the two stimuli, under five assumptions about what drives the third muscle. (e) Closed-loop search in silico for an "animal" whose love spot lies 20° higher than the model assumes (dark: tracking noise of 15 % of the effect; light: 38 %; grey: search started from an arbitrary stimulus). (f) Events of one glance of a simulated event sensor.

P1 MESH: electrical synapses at connectome scale (Fig. B1a). MESH finds every location where the membranes of two selected FlyWire neurons lie within 100 nm, groups the contacts into sites, validates them in the 8-nm EM volume (only membrane appositions without a chemical specialisation are kept) and serves them to experts in a viewer. For the 26 HS, VS and retinal motor neurons (RMNs) of both hemispheres it found 619 sites; 49 motor-neuron–LPTC sites survived validation [14,41].

P2 The electrical gaze pathway. Each retinal muscle has one motor neuron (MOS RMN: FlyWire type CB0901; MOT RMN: CB0804), and the two muscles pull orthogonally (51° and 320°). Neurobiotin fills and MESH agree that the MOS RMN is coupled to the three HS cells and VS1–4, and the MOT RMN to the HS cells only; a linear sum over the coupled cells predicts each RMN's preferred direction within 8°. Mutants lacking the gap-junction protein shakB show a reduced reflex, and spontaneous retinal movements are preceded by motor-related potentials in HS cells that mirror the visual consequence of the movement ($R^2 = 0.97$, slope −1) [14].

P3 The model, validated on *Drosophila* (Fig. B1b). One pipeline runs from stimulus to eye movement: an eye map with local lattice axes [109], motion detectors with multilevel opponency [26], LPTC receptive fields, a Hodgkin–Huxley network of electrically coupled HS and VS cells [11,32] and spiking motor neurons ported from MESH, and the muscle model of [14]. Two parameters are fitted, the rotation of the holder frame and one wiring ratio. The model reproduces the preferred directions of the MOS (212° against 212°) and MOT (146° against 141°) motor neurons and the direction of the optokinetic reflex (mean error 17°; [14]: 7°), and it predicts VS1–4 → MOS junctions twice as strong as HS → MOS, a number that T1.1 reads off the connectome.

P4 Two stimuli, three answers (Fig. B1c,d). The hoverfly model has both eyes, four HS cells with the male HSN narrowed to the bright zone [1], and a third muscle driven by HSN. Stimuli are square-wave gratings (wavelength 20°, 30°/s) on a display of 180° × 125° [22]. A full-field grating does not match the sexes (difference ≥ 23 % in every direction), because the coupled HS network carries wide-field motion to the muscle's motor neuron. Sparing the love spot does: with a 17° mask and downward drift the male-to-female ratio is 1.05, with a 12° mask 1.00. The grating inside the love spot gives 2.9 for upward drift

(third-muscle motor neuron: +13 Hz) and at least 2.4 in every direction; the narrowed HSN alone, with the muscle cut, gives 1.2. The same two stimuli separate the alternatives (Fig. B1d): drive by horizontal cells of any kind leaves the matched stimulus near 1 (1.0–1.2) and the love-spot stimulus above 2; drive by the vertical cells reverses the pattern (1.9 and 1.3); no visual drive gives 1.0 and 1.2. Which horizontal cell drives the muscle is then read from dye fills and from the direction tuning of the muscle.

P5 The animal in the loop (Fig. B1e). A search that reads only the retinal movement can correct the model. In silico, the "animal" was a male whose love spot lies 20° higher than the model assumes and whose muscle pulls in another direction. The objective was the movement that the model without the third muscle does not explain, read with noise of 15 % or 38 % of the effect; an evolution strategy proposed nine stimuli per generation. Started from the model's stimulus, four simulated sessions enlarged the patch, moved it up to 15° toward the true love spot and raised the muscle-specific movement by 18–33 % within about 100 presentations, five minutes of experiment, and by 26–37 % within 180. Started from an arbitrary stimulus, the search reached less than a third of the model-designed effect in the same time: the model is needed to start the search, the animal to finish it.

P6 An event camera with a moving sensor (Fig. B1f; Part I, Fig. 2c). Simulated scenes are static dead-leaves images with 1/f texture; the sensor has 128 × 128 pixels, lens blur, a contrast threshold of 20 % (± 3 % between pixels) and noise events. In one glance the sensor sweeps by a given distance while a dark 2-pixel target moves 2 pixels. Registering the scene between two glances to a quarter of a pixel needs a movement of at least half a pixel and works best at 2–6 pixels; the target, found by a size-tuned detector with lateral inhibition [61], is lost beyond one pixel. This holds for sparser clutter and faster targets; a 4-pixel target tolerates more movement, a low-contrast target less, and a 1-pixel target stays below threshold. The number of events, the currency of energy on neuromorphic hardware, grows eight-fold from half a pixel to two pixels. These numbers define the movement policies of T4.1.

P7 Comparative anatomy. In our survey of 167 insect species, hoverflies are among the earliest-branching flies with a specific retinal muscle and an optokinetic reflex. Both known muscles probably derive from antennal musculature [2]: the first clue to where the third comes from.

P8 Edge AI, teaching and participation. TinyML classifiers for real-time event detection on Arduino-class hardware (Computational SpikerBox, SpikerBot; NIH SBIR), a participatory study with pupils as first authors [88], and IU computer-vision courses from [semester; fill] with [n] students.

2. Methodology and work plan

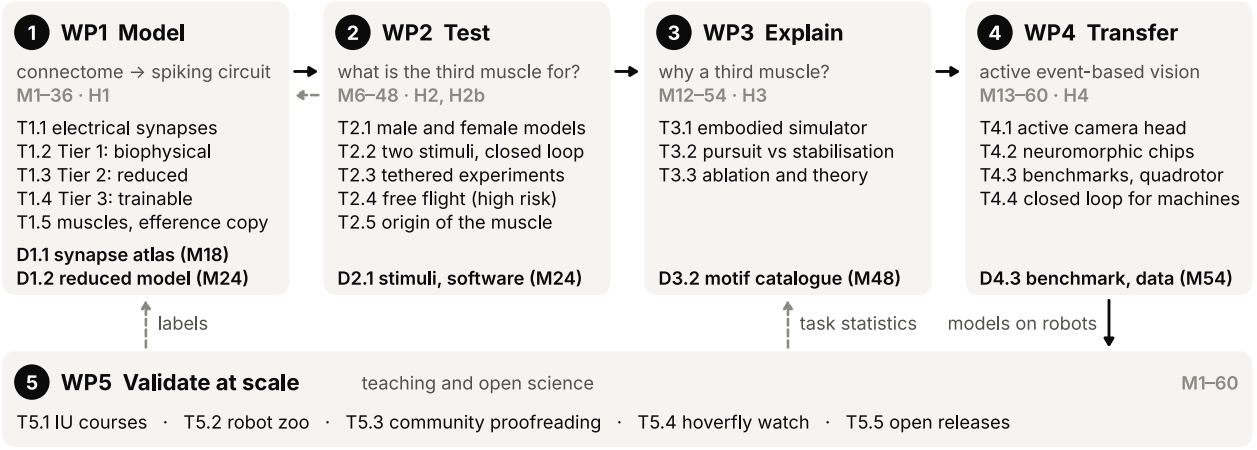


Figure B2. Work packages, tasks, deliverables (D), milestones (MS) and the main dependencies. Solid arrows: models and motifs flow from WP1 through WP3 to WP4 and onto robots; dashed arrows: data, labels and task statistics flow back from WP2 and WP5.

Five work packages share one modelling ladder and one stimulus engine (Fig. B2). "→ H" marks the hypothesis a task tests; D = deliverable; M = project month. Team (§6): PD1, PhD 1, PhD 2 and a research engineer (RE) at IU; PD2 at the MPI for Biological Intelligence; PhD 3 at LMU; student assistants (SA).

WP1 Connectome → spiking circuit, with electrical synapses (M1–36; PhD 1, PD1, PI) → H1

- **T1.1 Electrical synapses of the gaze system in both sexes.** Neuron set: all lobula-plate tangential cells (the HS, VS and the further types catalogued in [90]), the four RMNs, neck motor neurons and optic-flow descending neurons, in the female FlyWire connectome (v783 [12]) and the male central-nervous-system connectome (v1.0 [39]). MESH detects appositions; a 3D convolutional classifier trained on the expert-validated sites (membrane apposition, pentalaminar profile, no chemical speciali-

sation; negatives are rejected sites) ranks the rest; uncertainty-ranked queues go to expert and community proofreading (T5.3). *Validation:* held-out expert labels (precision and recall), agreement with the neurobiotin coupling in [14] and with the innexin immunostaining maps of [50]. *Sex comparison:* cell types are matched across the two connectomes with the cross-dataset typing of [47,39]; chemical and electrical connectivity of the gaze circuit are compared type by type. The expected result, a monomorphic circuit in *Drosophila*, is the baseline against which the hoverfly dimorphism is read; a difference would itself be a finding for H1's corollary. **D1.1 (M18):** open atlas of candidate electrical synapses of the fly gaze system, female and male, with the first sex comparison of this circuit (corollary of H1).

- **T1.2 Tier 1, biophysical.** Morphologically reduced multi-compartment HS, VS and RMN models built from the EM skeletons, active spike generation in the RMNs, chemical synapses from the connectome and gap-junction conductances proportional to validated apposition area. Parameters are fitted by gradient descent in a differentiable simulator [66] to published HS/VS physiology [11,31,32], RMN calcium tuning and the optogenetic displacement vectors of [14]. The front end is the detector front end of the model (P3), with the connectome-constrained optic-lobe model of [45] as an alternative input.
- **T1.3 Tier 2, reduced.** Integrate-and-fire or adaptive-exponential point neurons with linear gap-junction coupling. Effective couplings come from Tier-1 transfer impedances. For each module (dendritic integration, gap-junction transfer, muscle mechanics) small Kolmogorov–Arnold networks [93] are trained on Tier-1 input–output pairs, pruned and regressed to closed-form expressions, so that the reduction is explicit and inspectable. We report error bounds on RMN tuning and latency (target < 10 % against Tier 1). **D1.2 (M24):** reduced model and the reduction toolkit, reusable for other circuits.
- **T1.4 Tier 3, trainable.** A connectome-masked SNN with sign constraints, trained with surrogate gradients [64,65] in a JAX/PyTorch stack that exports to the neuromorphic toolchains used in WP4. Gap junctions enter as differentiable linear couplings, which also regularises training.
- **T1.5 Motor readout and efference copy.** Muscle activation → force (Hill-type) → retinal displacement through the measured pull directions; the efference copy is a model output, tested against the HS recordings in [14]. Validated against retinal tracking in WP2.

WP2 What is the third muscle for? (M6–48; PD2 at MPI-BI, PhD 3 at LMU, PI) → H2, H2b

- **T2.1 Model family.** Male and female circuits are built by homology (HSN, HSE, HSS, HSNE, VS; motor neurons of MOT, MOS and the third muscle, MIO), for both eyes, constrained by the tomography atlas (P7), the fills of T2.3 and published hoverfly physiology [1,15,17,23]. The male models differ in what drives the MIO (HSN as in H2, HSNE, pooled HS, pooled VS, a small-target input [16,63], or no visual drive), in its pull direction (twelve directions) and in whether its two limbs share one motor command.
- **T2.2 Stimulus design and closed loop.** (a) The matched and the love-spot stimulus (P4) and, for every pair of male models, the stimulus on which they disagree most [30], within the limits of the display and under a naturalness prior. The expected evidence per trial sets the number of animals: the **power analysis is done in silico before any experiment**. (b) The closed-loop search (P5) as real-time software: tracking of both retinae, an optimiser that proposes the next stimulus (evolution strategies, Bayesian optimisation [110,111]) and the model's stimuli as starting points. (c) A gaze-contingent mode in which the pattern moves with the tracked retina, so that the visual feedback of an eye movement can be removed, restored or reversed. **D2.1 (M24):** stimulus set and closed-loop software, open.
- **T2.3 Experiments (MPI for Biological Intelligence, Fenk laboratory; PD2, M7–42).** Tethered male and female *Eristalis* in front of a bowl-shaped projector display [22]; retinal tracking of both eyes through the cornea by deep-pseudopupil imaging under infrared light [7,14]; electromyography of MOT, MOS and MIO (the MIO is identified by its position between the eyes [2]); intracellular recordings and neurobiotin fills of HSN, HSNE and the RMNs with sharp electrodes, the standard method in hoverfly tangential cells [1]; section of the MIO through the dorsal cuticle as the causal test (prediction: the male then responds to the love-spot stimulus like the model with the muscle cut); animals reared year-round [19]. *Analysis:* hierarchical Bayesian comparison of the model family across animals [35].
- **T2.4 Free-flight retinal tracking (high risk, high gain).** Pursuit of artificial targets in an indoor arena [21] with head-mounted ultralight markers and high-speed cameras. Started only after T2.3 succeeds (decision at MS4); no other task depends on it.
- **T2.5 Developmental and genetic origin of the MIO (LMU, R. Merrill; PhD 3, M7–48) → H2b.**
  - *Origin:* a developmental series of male and female pupae (micro-CT, phalloidin confocal imaging) identifies the anlage the MIO forms from: a split of the MOT anlage, or a further repurposed anten-

nal muscle, as proposed for MOS and MOT [2].

- *Genes:* sex- and stage-resolved RNA-seq of the anlage region at three pupal stages (pooled, low-input protocols) with the chromosome-level *E. tenax* genome [44]; male-biased muscle-identity genes and the sex-determination factors *doublesex* and *fruitless*; candidates localised by in situ hybridisation chain reaction in the muscle and in its motor neuron.
- *H2b, the motor neuron specifies the muscle:* the male-specific Muscle of Lawrence of *Drosophila* forms only when its motor neurons carry the male *fruitless* programme [42,43,79]. We predict the same logic here: a male-specific identity of the MIO motor neuron induces the muscle.
- *Comparison:* syrphid species with and without a male MIO test whether candidate changes track the trait. *Exploratory, high risk:* RNAi knock-down in pupae.

WP3 Theory: why a third muscle? (M12–54; PD1, PI) → H3

- **T3.1 Embodied simulator.** Compound-eye sampling (including the male bright-zone resolution [15]), retinal actuators, simplified flight dynamics; GPU-based and open.
- **T3.2 Task optimisation.** Tier-3 networks are trained on *pursuit* (small fast targets in clutter [16,21]) or on *stabilisation* (hovering against optic flow; feeding approach [18]). Both conditions share the wiring prior and the spike-count energy penalty [80], and both may recruit a third actuator and shape its movement policy. Readouts: receptive-field width and position of HSN-like units, recruitment and use of the third actuator (compared with the measurements of T2.3) and the performance–energy trade-off. Controls: shuffled wiring prior, a range of energy penalties, ≥ 10 seeds, and a variant in which the sensor-movement policy is optimised jointly with the network.
- **T3.3 Analysis.** Ablation to the minimal motif set; linear-response and mean-field theory of gap-junction-coupled populations; information-theoretic receptive-field analysis; Kolmogorov–Arnold readouts that convert trained networks into equations. **D3.2 (M48):** the motif catalogue.

WP4 Active event-based machine vision (M13–60; PhD 2, RE, PD1, PI) → H4

- **T4.1 Active event-camera head.** A wide-angle and a telephoto event camera, the machine counterparts of eye and bright zone, each on a two-axis piezo tip-tilt stage with sub-pixel steps, on a pan-tilt gimbal and a motion platform that replays ego-motion. Movement policies: none, fixed artificial microsaccades [54], fly-derived (recorded in WP2), task-optimal (P6) and learned (T3.2); every motor command is available to the network as an efference copy. Measures: event yield, scene registration, target detection and latency.
- **T4.2 Motif SNNs on neuromorphic hardware.** Catalogue motifs (D3.2) become event-driven SNNs, deployed on a sensor-integrated chip with sub-milliwatt resting power [70] and on large-scale systems [67,68] through EBRAINS and vendor research programmes. Ablations: a Kolmogorov–Arnold parameterised event front end against convolutional front ends.
- **T4.3 Benchmarks (civil tasks, pre-registered protocols).**
  - **B-I Small-target detection and tracking in clutter:** our own *HoverEye* dataset, recorded with the active head (balls, insects and small drones against vegetation, with motion-capture ground truth) plus the public EV-IMO set [60]. Baselines: insect-inspired target detectors [61,107], event-based deep detectors [94] and converted SNNs. Metrics: detection F1, track continuity, event-to-output latency.
  - **B-II Ego-motion and optic-flow estimation:** MVSEC [59] and DSEC [58]. Baselines: E-RAFT [57], spiking optic-flow networks [55,56] and the fly-derived velocity CNN [38]. Metrics: endpoint error, latency, synaptic operations and measured energy on hardware against a GPU board.
  - **B-III Closed-loop hovering and pursuit on a quadrotor:** an indoor event-camera quadrotor with an on-board neuromorphic processor in a netted motion-capture arena, the robotic counterpart of the hoverfly arena of [21]. Tasks: hold position against optic flow (female task) and intercept a moving target (male task). Baselines: classical flow-based control and a GPU CNN pipeline [55]. Metrics: position error, time to capture, energy per trial.

  Protocol. Each benchmark is pre-registered (tasks, splits, metrics, baselines, seeds, stopping rules). Models are matched in parameter count and in synaptic operations per inference; energy is measured with a power meter on the deployed hardware and reported as accuracy per joule next to the GPU reference; five seeds per model. H4 counts as supported if the motif SNNs with active sampling reach at least the baseline accuracy at one third of the energy, or 20 % higher accuracy per joule, on two of the three benchmarks. **D4.3 (M54):** the benchmark, HoverEye and the `hover-snn` library.

- **T4.4 Closed loop for machines.** The WP2 engine tunes the movement policies on the hardware, with task performance as the objective, and generates controversial stimuli for CNN–SNN pairs as a robustness diagnostic (H4d).

WP5 Embodied validation at scale, teaching and open science (M1–60; PI, SA, IU courses)

- **T5.1 Course integration.** HOVER modules in IU computer-vision courses every semester; student projects deliver benchmark entries and robot data; distance students receive a robot kit by mail.
- **T5.2 Robot zoo.** Schools, makerspaces and citizens run fly-type, female-type and male-type circuits on low-cost two-wheeled camera robots over printable pursuit and stabilisation courses. Logs go to an open database and are analysed with mixed-effects models [35] with participant and device as random effects: thousands of embodied trials under uncontrolled optics and lighting.
- **T5.3 Community proofreading** of candidate electrical synapses in the MESH viewer, with an expert gold standard, feeding T1.1 [85].
- **T5.4 Hoverfly watch.** Smartphone slow-motion videos of male hovering and pursuit in gardens, stripped of location data, give natural pursuit statistics for T3.2 and B-I.
- **T5.5 Open releases and evaluation.** Code under MIT, data under CC-BY, teacher kits; firmware hardware-neutral (one commercial, one open platform); learning outcomes measured before and after.

Conflict of interest: the PI holds a senior role at Backyard Brains, the manufacturer of one of the candidate classroom robots. IU procurement selects robots against a hardware-neutral specification without the PI's involvement, the quadrotor and the active head (WP4) use unrelated vendors, and all code runs on at least one open platform.

3. Timeline, milestones and decision points

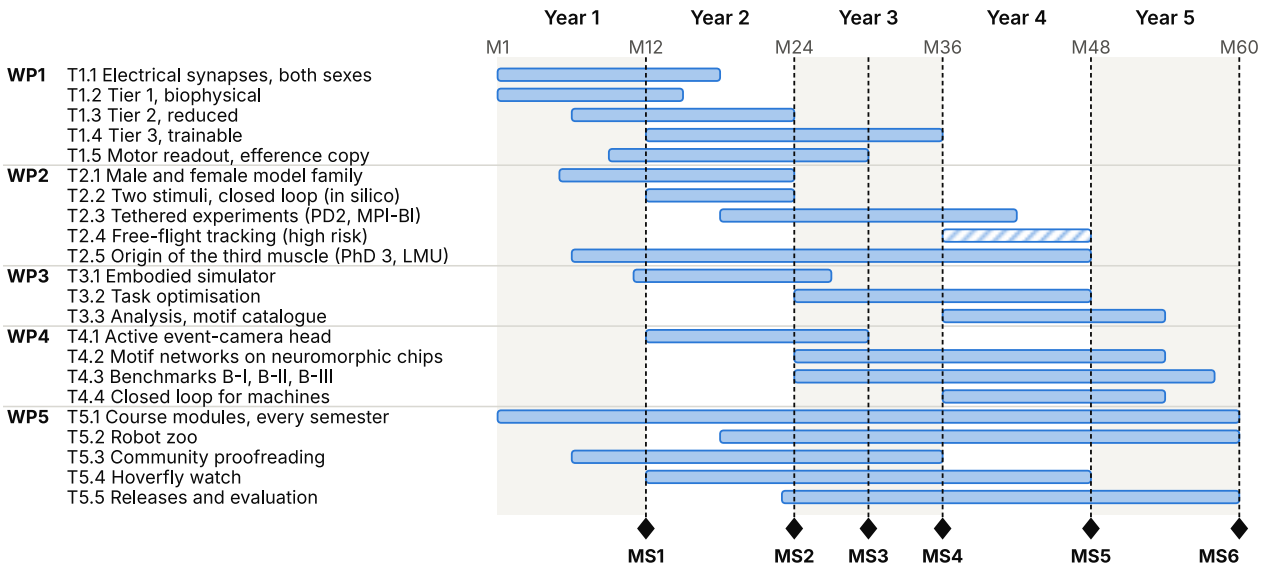


Figure B3. Gantt chart of the 22 tasks and six milestones (M1 = September 2028, placeholder). WP1, WP3 and WP4 form the modelling and engineering spine; WP2 and WP5 run alongside and feed data back.

MS	Month	Milestone	Decision
MS1	12	Electrical-synapse atlas v1 (female); Tier-2 *Drosophila* model	Tier-2 error against Tier 1 < 10 %? If not, Tier 1 serves WP2
MS2	24	Male atlas and sex comparison; male and female *Eristalis* model families; two stimuli and closed-loop software ready	Do the stimuli separate the models with the planned number of animals? If not, refine models before experiments
MS3	30	Active head and quadrotor flying; robot zoo launched; HoverEye v1	Hardware access confirmed, else GPU emulation with energy modelling
MS4	36	H1 decided; tethered *Eristalis* data complete	Constraints fixed for WP3; go/no-go for T2.4
MS5	48	H2, H2b and H3 decided; motif catalogue	Motifs passed to WP4
MS6	60	H4 decided; benchmark, models, dataset and kits released	—

4. Hypotheses: tests and what each outcome teaches

Hyp.	Supported if	Falsified if	Value if falsified
H1	The Tier-2 SNN with EM-derived weights predicts RMN preferred directions (± 15°), eye-movement direction and latency in both sexes	Connectivity-only models miss ≥ 2 RMN tunings	Unannotated or neuromodulatory inputs matter; the MESH search is widened
H2	The matched stimulus moves male and female retinae alike; the love-spot stimulus moves the male retina more, the MIO fires with motion in the love spot, and the difference disappears when the MIO is cut	The sexes differ on wide-field motion (an amplifier of the reflex), or the MIO does not follow visual motion (spontaneous use)	Either outcome still says what the muscle is for; the closed loop finds its best stimulus without the model
H2b	*fruitless* or *doublesex* are expressed male-specifically in the MIO motor neuron and anlage; the muscle forms from a defined anlage only in males	No sex-biased expression in the motor neuron; specification is muscle-intrinsic	A muscle-intrinsic programme; still the first developmental account of a dimorphic eye muscle
H3	Narrowed dorso-frontal integration and an extra effector emerge only under the pursuit objective	They emerge under both objectives, or neither	Points to developmental or genetic constraint rather than task optimality
H4	Motif SNNs with active micro-movements beat matched baselines in accuracy per energy on ≥ 2 of 3 benchmarks	No advantage	A rigorous, open negative result for bio-inspired claims; the benchmark remains
H1b	The Tier-1/2 model reproduces, with no added pathway, the motor-related potentials of HS cells during spontaneous retinal movements (direction tuning $R^2 \geq 0.9$ against [14])	The coupling alone cannot produce them	Identifies the missing element (a chemical pathway, a delay) and constrains Tier 1
H2c	Spontaneous MIO activity produces motor-related potentials in the male HSN; the female HSN shows none	HSN motor-related potentials are the same in both sexes	The MIO channel is one-way; the efference copy is not part of the dimorphism
H3b	The narrowed male HSN field emerges only with the spike-count penalty	It emerges at any penalty, or never	Separates task from efficiency as the cause of the narrowing

Computer-vision hypotheses tested in WP4 (all against baselines matched in parameters and synaptic operations; metrics: accuracy per joule and latency; pre-registered):

	Prediction	Test
H4a Task-specific sampling	Pursuit-type micro-movement policies beat fixed microsaccades [54] on small-target detection at equal event rate; stabilising policies win on ego-motion; no fixed policy wins both	B-I and B-II with the four policies of T4.1
H4b Efference-copy cancellation	Feeding the motor command back into the motion units through the LPTC–RMN coupling motif removes self-generated events from the flow estimate better than software compensation [54]	Flow error during sensor motion, B-II and B-III
H4c Electrical pooling	Linear, delay-free pooling of local motion into wide-field units reaches a given ego-motion accuracy at lower event-to-command latency than learned synaptic pooling	Latency at matched accuracy, B-II and B-III
H4d Opponency and normalisation	Push-pull coding with dynamic signal compression [38] is more robust to contrast change and to controversial stimuli than convolutional front ends	T4.4 diagnostic on B-I and B-II
H4e Specialisation per joule	Under one energy budget, two task-specialised circuits with a shared front end, the dimorphic solution, beat one generalist circuit on the combined pursuit and stabilisation task	B-III with pursuit and hover episodes mixed
H4f Hyperacuity	Known micro-movements let the network localise small targets below the pixel pitch, as photoreceptor micro-movements do in the fly [51]	Localisation error against pixel size, B-I
H4g A second movement for a special zone	A telephoto "bright-zone" sensor with its own micro-movement, added to a wide-angle sensor, beats one moving sensor of equal pixel count and energy on the combined pursuit and stabilisation task	B-III with the two-camera head of T4.1

5. Risks and mitigation

Risk	Likelihood	Impact	Mitigation
Gap-junction classifier does not generalise beyond validated sites	Medium	Medium	Expert and community proofreading; gap-junction weights become bounded free parameters
No *Eristalis* connectome, so the homology mapping is wrong	Medium	High	Competing model families instead of one model; discriminating stimuli; fills as ground truth
Hoverfly physiology slower than planned; MIO hard to reach for electromyography or section	Medium	High	An experienced postdoc at the only site with the set-ups; the two stimuli and the closed loop need only retinal tracking; WP1, WP3 and WP4 do not wait
Developing muscle tissue too small for RNA-seq	Medium	Medium	Pooled samples, low-input protocols; HCR for *Drosophila*-known candidates as fallback
Free-flight tracking fails	High	Medium	Tethered closed-loop virtual flight; no task depends on T2.4
Quadrotor integration delayed or unsafe; hardware access restricted	Medium	Medium	B-I and B-II are independent; the active head covers egomotion; indoor netted arena only; two hardware routes plus GPU emulation with calibrated energy models
SNNs lose to deep baselines (H4 null)	Medium	Medium	Reported rigorously; hybrid motif front end with learned back end as a secondary result
Recruitment delays	Low	Medium	Posting on EURAXESS at signature; IU computer-vision students as the MSc → PhD pipeline

6. Team, working arrangements and environment

- **PI (50 %, above the 40 % minimum):** scientific lead; WP1 and WP3 modelling; WP5 lead.
- **Team, three beneficiaries.** At IU: PD1, computational neuroscience and SNN theory (48 months); PhD 1, connectomics and machine learning; PhD 2, event-based spiking vision and neuromorphic hardware; a part-time research engineer for the active head, the quadrotor and the robot fleet (M13–48); student assistants. At MPI for Biological Intelligence (Active Sensing, L. Fenk): PD2, hoverfly physiology (36 months, M7–42). At LMU Munich (R. Merrill): PhD 3, evolutionary developmental genetics (48 months). The PI selects, funds and supervises all team members; the partner group leaders co-supervise locally and give access to their set-ups.
- **Relation to the Fenk laboratory's ERC project** on *Drosophila* eye-movement mechanisms: HOVER works on the hoverfly's sex-specific muscle, on models and on machines. Binocular coordination of the retinae and its role in depth perception [7] remain the Fenk laboratory's topic and are not part of HOVER. The division of scope is fixed in a written collaboration agreement; data and tools flow both ways.
- **Doctoral route:** PhD students enrol at LMU (doctorate-granting), co-supervised by the PI [confirm the agreement and whether IU has doctoral rights].
- **Collaborators** (written consent before submission; roles limited to what is stated): L. Fenk (physiology set-ups, *Drosophila* RMN data, tomography); R. Merrill (genomics, developmental genetics); [neuromorphic hardware contact]; [event-camera or drone partner].
- **Working arrangements:** monthly all-team meeting in Munich; quarterly WP reviews against the milestones; one shared repository with continuous integration; a data management plan by M6.
- **Data and software management:** every model tier, stimulus set, recording and benchmark run is versioned and released with a DOI (Zenodo, DANDI for physiology, the FlyWire and MaleCNS annotation channels for proofread synapses); PhD 1 is data steward (10 % of time); every milestone review includes a FAIR audit.
- **Environment:** the IU computer-vision group, students and GPU resources [confirm]; physiology and tomography at MPI-BI; genomics at LMU; EBRAINS neuromorphic access.

7. Ethics, civil focus, open science and data

- **Civil focus:** HOVER has an exclusive focus on civil applications. The quadrotor flies indoors only. Insects only (no vertebrates). Human participants (students, citizens including school pupils) take part with informed and parental consent under IU ethics approval, with GDPR-compliant data handling; uploaded videos are stripped of location and metadata.
- **AI:** HOVER develops AI-based vision models; no personal data are used for training.
- **Open science:** immediate open access to publications; code, data and protocols as in T5.5.

Funding ID (not counted in the page limit)

Required: current research grants and their subject, and any ongoing application for work related to this proposal (IfA p. 29).

Title	Funder	Amount	Period	Role	Status	Relation to HOVER
[IU start-up package]	IU	[€]	[ ]	PI	[cur-rent]	[what it pays for; no overlap with HOVER costs]
[LMUexcellence Innovation Fund project]	LMU	[€]	[2024–]	[role]	[status]	[subject; no overlap]
[other current grants]						

No applications for work related to HOVER are pending.

References

See `references.md` (not counted in the page limit).

References

Numbering 1–44 follows the DFG proposal so that figure and text citations carry over; 45 onwards were added for the ERC version. Applicant publications are in **bold**. Entries marked [verify] could not be checked against the publisher's page from this environment; check them before submission. Author lists must appear exactly as published (IfA p. 28).

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